Analyzing H-1B Visa Sponsorship Trends in Finance-Related Occupations (2016-2024) Using Text Mining

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Introduction

The United States remains the top destination for people across the world, whether for shelter, education, or job opportunities, as immigrants or nonimmigrants. For many, economic reasons are the deciding factor. The U.S. government has instituted a legal pathway for those seeking employment in the country, namely the H-1B visa program, which permits entry to nonimmigrant workers that have specialized expertise from abroad (DOL 2025). The initiative was designed to fill in gaps in the American labor supply, as its economy continues to expand in which many industries, ranging from technology to commerce to manufacturing, are in need of a high-skilled workforce. Currently, it has an annual statutory cap of 65,000 visas, plus an additional 20,000 visas for those who have recently earned a graduate degree from a U.S. institution (American Immigration Council 2025). In 2024, around 400,000 people were approved with H-1B visas and it is estimated that H-1B holders make up roughly 730,000 of U.S. residents (Im et al 2025; FWD.us 2025). About 90% of petitions are for positions requiring STEM knowledge, according to a Pew Research study; among these, computer, engineering, math and science are the top four fields (Ruiz 2017). Simultaneously, finance is one field that comprises a significant portion of the U.S. economy; together with the insurance industry, it makes up 7.3% of the total GDP and employs more than 6.7 million people across the country (ITA n.d.). Over the years, finance has increasingly been reliant on roles that require STEM expertise, like data analysis, which many H-1B applicants likely apply for. However, there has been little research on financial H-1B jobs specifically, as most of the existing literature tends to focus on explicitly STEM fields or on the H-1B program as a whole. When it comes to government data, they tend to combine both aforementioned industries into one group, making it difficult to conduct research on just finance, which is further discussed below. H-1B visa approval rates can also be reflected by fluctuations in

labor supply and demand, as well as changes in immigration policies. To explore these patterns, this project applies natural language processing (NLP) techniques using a transformer-based model known as BERT to quantify the number of H-1B petitions in finance-related occupations, classify them into specific subfields based on their associated job titles, and analyze how demand for foreign labor has shifted over time from 2016 to 2024 across these groups. The following sections outline recent policy changes to the H-1B program, describe how the data was collected and analyzed, and present the main findings, including some possible explanations for the trends.

Background

The H-1B visa program was first created in 1990 as part of the Immigration Act of that year to allow foreign workers to work in specialty occupations, defined as jobs that require a bachelor's degree and/or professional license in a specialized field, like engineering, science and academia (Batalova 2010). From an economic standpoint, H-1B workers represent a comparative advantage for the U.S. economy relative to that of the countries they emigrate from, because such inflow of human capital also brings along technical expertise and innovative knowledge. In domestic terms, there is limited evidence showing that H-1B workers are hired due to firms' incentives to minimize costs. One study finds that the H-1B program does not have any adverse effect on employment outcomes and wages for native workers (Kerr and Lincoln 2010). Another by Peri et al. (2014) finds that the presence of foreign STEM workers has led to an increase in wage for domestic workers in the same field, as well. The American Immigration Council (2025) also argues that foreign workers help boost consumer demand and encourage business growth, on top of being additional sources for tax revenues for the U.S. government.

Nonetheless, like immigration, the H-1B program is subject to contentious political debate as the U.S. economy has experienced volatility over the past decades and the labor market has

become more competitive. As stated earlier, the cap for H-1B sits at 65,000 per year, but it used to be almost doubled this number back in 1999 and early 2000s to make up for severe labor shortages, until it got reverted back to 65,000 in 2004 (Batalova 2010). Since then, the political atmosphere surrounding immigration has intensified beginning with Trump's first presidential term in 2016, which continues throughout the subsequent term under Biden that effectively lasted until the end of 2024. Both administrations adopted different policies and agendas toward this issue, at least rhetorically, as America has experienced major shocks to its economy (i.e., global pandemic, etc.) and a deepening political divide across its society.

In April 2017, President Trump signed the Buy American and Hire American Executive Order, which tightened immigration enforcement and made visa programs like H-1B more restrictive (USCIS 2021), which reduced the size of the applicant pool, as only the highest-paid and most-skilled applicants would be considered for admission. In October 2020, the Trump administration made the process even harder by raising the minimum wage employers had to offer to these foreign workers and narrowing the definition of a specialty occupation so that fewer people would qualify (Anderson 2020). This led to a surge in visa denials, but after President Biden took office in 2021, his administration reversed some of these changes, bringing denial rates down significantly compared to the levels seen during Trump's term (Immigration Policy Tracking Project 2021).

However, in late 2023, the Biden administration introduced a new rule similar to Trump's previous policy by curtailing which jobs could count as specialty occupations (Anderson 2023). Nearing the end of his term, Biden made some final changes that granted more flexibility with starting dates for work, which also practically extended work authorization for F-1 students who may seek a new H-1B visa; additionally, Biden's new rules brought back deference to prior

approvals, but they still kept the same strict wording as before regarding specialty occupations (Shie et al. 2024). The COVID-19 pandemic from 2020 to 2023 also caused a major disruption to travel and various economic activities, resulting in freezes in new hiring and a likely decline in H-1B visa approvals.

Methodology

Data Collection

This project mainly uses data from the U.S. Citizenship and Immigration Services (USCIS) and the Department of Labor (DOL), as they are both official and publicly available. The files came with CSV and XLSX formats. The USCIS website contains the H-1B employer data across all U.S. states from fiscal year 2009 through fiscal year 2024, during which employers submitted petitions to employ H-1B nonimmigrant workers (USCIS 2024). It includes information about the approval and denial of H-1B visas, arranged into different sectors as defined by the NAICS code; here, finance is categorized in Group 52, which combines both finance and insurance industries. This classification system groups petitions by the type of firm they are applied under rather than their specific occupations, many of which may overlap with one another across sectors. As such, the approach was to isolate and identify any role related to finance based on job titles, irrespective of their industry codes; this process will be detailed in the section below. The DOL performance data contain records for all types of visas, but for this project, only data concerning LCA programs were considered because they include H-1B records. These LCA disclosure data are reported annually from 2016 to 2019, but starting from 2020, they are released every quarter. The files obtained from DOL are particularly voluminous, amounting to nearly 2.55 gigabytes in size, with each file containing hundreds of thousands of individual cases for each year. The datasets include many details for each case, such as case status, received and decision dates, job titles, employer

details, and wages. Although SOC occupational codes were also present in the datasets, there are finance-related roles that are classified under groups other than "13-0000 Business and Financial Operations Occupations" (BLS 2023). For example, financial clerks are classified under "43-0000 Office and Administrative Support Occupations," while securities and financial services sales agents fall under "41-0000 Sales and Related Occupations." Because of this, the study focuses primarily on case status and job titles alone, while the remaining variables were simply excluded.

Data Processing

All the following procedures were conducted using R within the RStudio environment. Before building any models, the data had to be cleaned and harmonized as they were unstructured for the most part. While the USCIS data were comparatively more structured and standardized, thereby requiring minimal cleaning, the DOL datasets presented such cases, as many cells appeared to have been completed in a fill-in-the-blanks manner, resulting in inconsistent labels and formats. For instance, when filtering by job titles related to finance, arbitrary words, irregular character strings, and some abbreviations frequently appeared, as shown in Figure 1.

JOB_TITLE
Financial and Investment Analysts - KBGFJG72284-11
Model/Analysis/Validation Senior Analyst
Accountants and Auditors - Kbgfjg179876-5
Accountants and Auditors - Kbgfjg140954-6
Director, Sponsor Underwriting, Capital Markets
Treasury Analyst
Accountants and Auditors - KBGFJG102001-7
Associate III
FINANCIAL RISK SPECIALIST
Assistant Vice President
Technical Data Analyst
Accountants and Auditors - KBGFJG265299-1
Tax Senior
Accountants and Auditors - Kbgfjg227420-4
Supervisor, Audit & Attest
Audit Manager
Accountants and Auditors - Kbgfjg159935-6
Accountants and Auditors - Kbgfjg182835-4
Accountants and Auditors - Kbgfjg202575-4
Accountants and Auditors - KBGFJG101848-6
Accountants and Auditors - KBGFJG132073-4
Lead Trade Operations Analyst
Accountants and Auditors - Kbgfjg131909-4
Associate
Accountants and Auditors - Kbgfjg87152-10

Figure 1. Example of entries for the job titles from one of the DOL H-1B disclosure data.

Filtering job titles by their textual components is essential to quantify the number of jobs related to finance. After importing each XLSX file into R, every column name in each file was standardized using the `janitor` package to prevent any potential inconsistency in formatting so that it would be feasible when consolidating the files into a single dataset. All the texts were normalized using the `dplyr` package, which includes converting the variable `case_status` to lower case, along with all its values (note that there are four types of case statuses, including "certified," "certified-withdrawn," "denied," or "withdrawn"). Cases without proper job titles were excluded. A custom keyword list was made that includes possible names for finance-specific roles that may appear in the datasets (Figure 2). Certain terms were entered in their root forms to take into account any variations in their spelling (e.g., combining "finance," "financing," and "financial" into just "financ"). This procedure ensured that each of the nine DOL datasets was

reduced to a manageable size to optimize code runtime, while also removing a large number of observations that were irrelevant to the research. These CSV files were converted to RDS format, as it is native to R, which is more efficient when attempting to generate BERT embeddings for a large number of observations.

```
finance_keywords <- c(
  "financ", "investor", "investing", "accounting", "accountant",
  "actuary", "treasury", "risk", "compliance", "quantitative",
  "portfolio", "equity", "asset", "securities", "hedge", "audit",
  "valuation", "credit", "bank", "underwriter", "tax"
)
```

Figure 2. Custom list of finance-related keywords used to filter H-1B job titles.

Embedding and Clustering Job Titles

As there are many subfields within the finance industry, sorting them into groups is a prerequisite to analyzing how each type of finance-related job has performed over the course of the sample period. Due to the unstructured nature of job titles in the datasets, grouping them using basic keyword matching was inadequate and at risk of misidentification, especially when many job titles refer to similar roles despite some differences in wording, thus requiring a more complex technique, in particular, NLP based on BERT. However, because this model is not supported in R and instead runs on Python, the `reticulate` package was needed to make Python run on RStudio. With the `sentence-transformers` package, the BERT model converted each individual job title into a numerical vector, which was then used to compare with other vectors in terms of similarities in semantics. Following Tiwari (2024), the `all-MiniLM-L6-v2` model was used, which maps short text into a 384-dimensional vector suitable for semantic search, clustering, and sentence similarity tasks. Before generating the embeddings, job titles were further cleaned by removing rank-related terms such as “SR,” “junior,” “president,” “director,” “associate,” and “CEO,” since the rank of the position was not relevant to the analysis. Any

numerical values and unnecessary punctuation were removed from those job titles. After further cleaning, the DOL datasets were combined into a single RDS file before being passed through the BERT model. The resulting embeddings were then saved in `dol_h1b_embeddings.rds`, while the list of unique job titles was saved separately in `dol_h1b_combined_titles.rds`.

Given the generated vectors, clusters were created using K-means to identify distinct branches of finance-related jobs, with the optimal number of clusters (k) determined using the silhouette score method. With the `sklearn.metrics` package, silhouette scores were tested for values of k from 5 to 25. Testing the method on the full dataset was initially tried, but it proved to be too computationally intensive, as the dataset contains more than 190,000 observations. Instead, the test was conducted using a sample of 1,000 randomly selected embeddings. As a result from Figure 3, the highest silhouette score was $k = 18$, which sat just under 0.080 (far from the ideal value of 1.0). Regardless, the overall scores were below even 0.10, with only small differences among them. As seen in the graph, $k = 8$ forms a clear local maximum at approximately 0.067 before the sharp increase beyond $k = 15$. For this reason, $k = 8$ was selected, as it allows for better interpretability and having a large number of clusters would only make the model more complex.

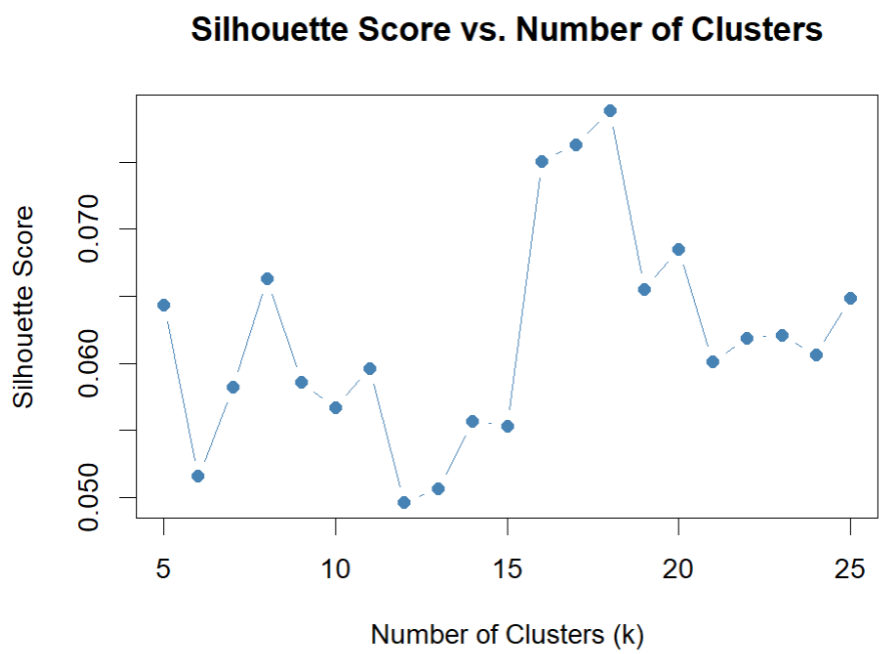

Figure 3. Plot showing the silhouette score for optimal number of clusters (*k*).

With $k = 8$ chosen as the number of groups, the job title embeddings were clustered using the K-means algorithm via `sklearn.cluster`. During this process, each job title was assigned to one of the eight clusters based their meaning similarities, with each one automatically labeled from 0 to 7. The cluster labels were added as a new column to each yearly DOL dataset and all yearly datasets were then combined into a CSV file. From here, the top fifty most frequent job titles within each cluster were extracted to determine how effective the clustering grouped similar roles together. Using these fifty representative job titles, one way to best visualize how each cluster is represented is through word clouds created using the `wordcloud` package, as this technique displays the most frequently occurring terms in which those with the highest frequency are presented in larger text size. The outputs of these word clouds are shown in Figure 4.

Figure 4. Panel of word clouds showing the most common job title keywords for each cluster.

Empirical Analysis

Cluster Classification and Interpretation

Based on the word cloud results illustrated above, the most prominent terms from each cluster are distinct enough to justify classifying each group into a separate subfield. Notably, there are broad words such as “finance” and “financial,” which also appear frequently across multiple clusters, particularly the first two, due to the fact that they are likely embedded within many job titles (e.g., “financial banking specialist” and “financial data analyst”). Both Clusters 0 and 1 share this issue, but there are several differences between them; the former includes terms like “banking” and “credit,” indicating roles related to banking and credit services, while the latter includes

"consultant," "services," and "specialist," thus representing roles focused on financial consulting. Cluster 2 is dominated by the term "risk," which naturally links to risk management, while Cluster 3 is associated with tax, audit and accounting. Clusters 4 and 5 share some similarities, as they both involve analytics; however, one is more related to applied business and financial analytics, whereas the other emphasizes quantitative research and evaluation. The final two clusters, Clusters 6 and 7, correspond to portfolio and asset management, and compliance and regulation, respectively. These classifications are summarized in Table 1 below.

Cluster	Finance Job Category
0	Banking and credit services
1	Financial consulting
2	Risk management
3	Tax, audit and accounting
4	Financial and business analytics
5	Quantitative research and evaluation
6	Portfolio and asset management
7	Compliance and regulation

Table 1. Labels for each cluster based on word cloud results.

Trends by Cluster and Case Status

There were 197,589 H-1B petitions across all clusters over the observed period, with an annual average of about 22,277. These petitions included all case statuses, as well as both new and continuing visa requests (either for initial entry or renewal). The total number of petitions increased progressively from 19,132 at the beginning of Trump's presidency up to 22,715 in 2019, when it fell back to the 2016 level by the end of his first term (Table 2).

	Years								
Cluster	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
0	1578	1675	1895	1863	1748	2931	2188	2338	2175
1	1336	1349	1517	1465	1209	1787	1279	1710	1762
2	1476	1553	2185	2212	2001	3141	2117	1869	1746
3	8468	8907	9372	9504	8189	11411	7973	10110	8051
4	3680	3737	4287	4097	3262	5569	3707	4222	3843
5	1504	1741	2041	2229	2133	3042	2092	2291	2157
6	511	506	650	616	567	1008	678	697	767
7	579	656	693	729	721	1034	784	795	774
Total	19,132	20,124	22,640	22,715	19,830	29,923	20,818	24,032	21,275

Table 2. Total number of H-1B petitions by cluster and year, including all case statuses.

The USCIS then saw a drastic increase to this number by 50.90% in 2021, which was during the COVID-19 pandemic, with 29,923 petitions received. This number returned to pre-pandemic levels between 2022 and 2024, during which it fluctuated between 20,000 and 25,000. Based on the figures from Table 2, Cluster 3 (tax, audit and accounting) represented the largest fraction of H-1B petitions every year, encompassing more than 40% of the total from 2016 through 2020, and in 2023. This was followed by Cluster 4 (financial and business analytics), which accounted for 16% to 19% of the shares, consistently making up just under half of Cluster 3's total each year. Cluster 0 (banking and credit services) ranked next, constituting 8.25% in 2016 and 10.22% by 2024. It was briefly overtaken by Cluster 2 (risk management) and Cluster 5 (quantitative research and evaluation) from 2017 to 2021, although its share was generally increasing over the longer period. Both Clusters 2 and 5 had comparable increases, but they became more prominent

beginning in 2018, as they fluctuate between roughly 7% and 11%. Cluster 1 (financial consulting) followed close behind, sitting at around 6.72% on average. The two smallest groups, Cluster 6 (portfolio and asset management) and Cluster 7 (compliance and regulation), had an average of about 2.97% and 3.3%, respectively. Among all these eight subgroups created by the embeddings, jobs related to assurance and accounting, as well as data analytics, were most popular for H-1B applicants, whereas those involving investment management, and compliance and regulations were not as high in demand.

Year	Certified	Certified-Withdrawn	Denied	Withdrawn
2016	16,700	1,270	508	654
2017	17,692	1,242	410	780
2018	20,188	1,229	485	738
2019	20,445	1,318	307	645
2020	18,859	378	257	336
2021	28,277	874	303	469
2022	19,354	975	170	319
2023	22,283	1,167	166	416
2024	19,705	1,007	240	323

Table 3. Total number of H-1B petitions by case status and year.

When examining by each case outcome instead, the approval rates remained consistently high over the years, accompanied by almost negligible denial rates (Table 3). The proportion of certified cases, excluding those that were later withdrawn, was lowest prior to 2020 (87% to 90%), nonetheless showed an upward trend. In 2020, the certified rate was at its peak (95.11%), but following that, it declined steadily to 92.61% by 2024, resembling a slight inverted-U shape. The

rates of denial were highest from 2016 to 2018, although they never surpassed 3%; in 2019, the rate dropped to around 1.35%, as it continued to gradually decrease following that, reaching as low as 0.69% in 2023. A similar pattern can be found in the withdrawal rates, despite remaining just above the denial rates by 0.39 to 1.84 percentage points. The certified-withdrawn rates were approximately 5-7% pre-2020 and fell to less than 2% in 2020—coinciding with the highest certified rate—before slowly climbing thereafter.

However, there were some variations across different clusters. Every cluster recorded a peak certified rate in 2020, while Cluster 1 trailed behind by one year. Moreover, all clusters saw a surge of at least 5% in that year, aside from Cluster 4, which had a more modest growth at 3.26%; among these, the largest increases were observed in Clusters 5 and 6, with both being greater than 8%. The aforementioned inverted-U pattern did not fully reflect the trajectories of certified rates in Clusters 6 and 7, as they kept oscillating between years, albeit with small margins, except for the uptick in 2020; for instance, in Cluster 7, the certified rate went from 84.97% to 84.45% and back up to 86.15% over the course of 2016 to 2018, a pattern that recurred in the subsequent years as well. Cases that were classified under Clusters 2 and 5 had among the lowest rejection rates across all years, with none exceeding 1.2% (e.g., dropping to as low as 0.16% and 0.22% in 2023, respectively). In 2024, with the exception of Cluster 0, every group experienced a minor increase in the denial rate. In spite of these disparities, every cluster showed a similar overall trajectory across all case statuses, pointing to the fact that these finance subfields had rather minimal impact on certification outcomes for H-1B petitions. Still, these distinctions showcased which types of finance-related roles were highly in demand in the job market, as evidenced by the volume of petitions per cluster.

At the same time, variations in aggregate sponsorship levels could be attributed more to macroeconomic shocks, particularly with respect to 2020 and 2021, when the COVID-19 pandemic was at its height. Excluding 2021 reveals that there were no significant differences in petition numbers across the two presidencies, as they stabilized around an average of 21,320 cases, deviating by around -10% to +13% from the mean. The surge of cases in 2021 may sound counterintuitive given that the pandemic had caused disruptions to movement, thereby limiting the amount of people being able to travel across countries for H-1B work. That said, this phenomenon can be explained by the 12.70% decrease in 2020, implying that such disruptions did occur, which caused a backlog of cases that were forced to be delayed to the following year (resulting in the spike), as many government offices were subject to lockdown. The slight decrease of 11.47% in total cases toward the end of Biden's term is also worth noting because it happened concurrently with changes to the H-1B eligibility criteria that were deemed as restrictive. Meanwhile, certification rates seemed to be closely tied to policy shifts under each administration. The rates of certified petitions were marginally higher during Biden's term, with at least 92% over each year, compared to 87-90% during Trump's term. The latter also oversaw relatively higher denial rates (mostly above 2%), which were roughly twice the rates observed under Biden.

The broader trends can be further corroborated by the USCIS data, showing all adjudicated H-1B petitions within the finance and insurance industry, as classified by NAICS, although the scope does not conform to the job title embedding data based on the DOL datasets. As shown in the graph below, there was a positive trend between 2018 and 2022 in terms of continuing approvals, whereas initial approvals remained mostly flat during the same period (Figure 5). The peak number of continuing approvals in 2022, similarly to what was discussed earlier, could be partly due to delays in the adjudication process at the beginning of the pandemic. The stark contrast

between the approval types showed that many of the certified petitions for the finance-related occupations in the main data may have consisted more of renewals than new filings. It is also plausible that employers rather chose to keep existing workers during times of economic volatility, especially when they were given temporary visas like H-1B. Likewise, the increase in initial approvals in 2024 (post-COVID-19) could mean that employers were able to sponsor new foreign workers as the economic conditions began to recover.

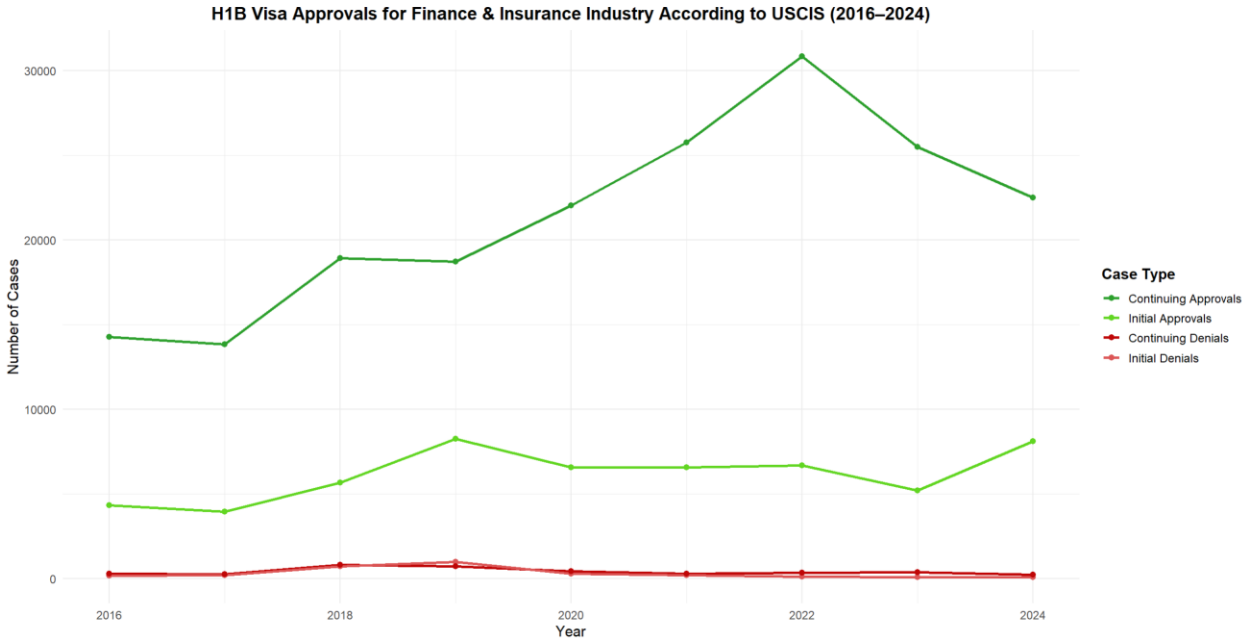

Figure 5. Plot showing the number of approvals and denials for H-1B visas in the finance and insurance industry (NAICS 52).

Limitations

There are several constraints throughout this research. Due to the massive size of the datasets, processing and transforming them, especially during the generation of the embeddings, require relatively extensive hardware resources. Some of these shortcomings are mitigated by utilizing a random sample of just 1,000 job title embeddings during the silhouette scoring process, as well as by converting the data in RDS format for more efficiency. Additionally,

because the methodology relies on job titles to create its own groupings from the ground up rather than standardized classifications (i.e., NAICS and SOC), some roles that are technically finance-related may have been excluded, while others may have been included despite not fully belonging to this intended group, for example, in cases where roles overlap with the insurance industry, given the close overlap between the two sectors.

Conclusion

This study has applied NLP using the neural model BERT, to measure the overarching patterns in the demand for skilled foreign labor in the U.S., based on H-1B petition filings, with a focus on finance-related occupations between 2016 and 2024. The clusters derived from the embeddings of job title descriptions (with $k = 8$) provide insights into which specialized branch of finance had the highest concentration (in this case, positions within tax, audit and accounting). The differences among these groups in terms of certification status were negligible overall, as these rates correlated more with changes in immigration policy during the two administrations of Trump and Biden. The petition volumes, in comparison, were relatively steady over the time period, but had a steep increase during the pandemic era, likely caused by the delays in administrative functions. Ultimately, this study offers an alternative approach in obtaining information from large, unstructured data. Further research may incorporate regression analysis to better interpret the causality of these findings and possibly expand the scope to other sectors of the U.S. economy to perform a cross-comparison.

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